

OPERATING SYSTEMS

Process Management and Scheduling

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OPERATING SYSTEMS

SRT - Shortest Remaining Time



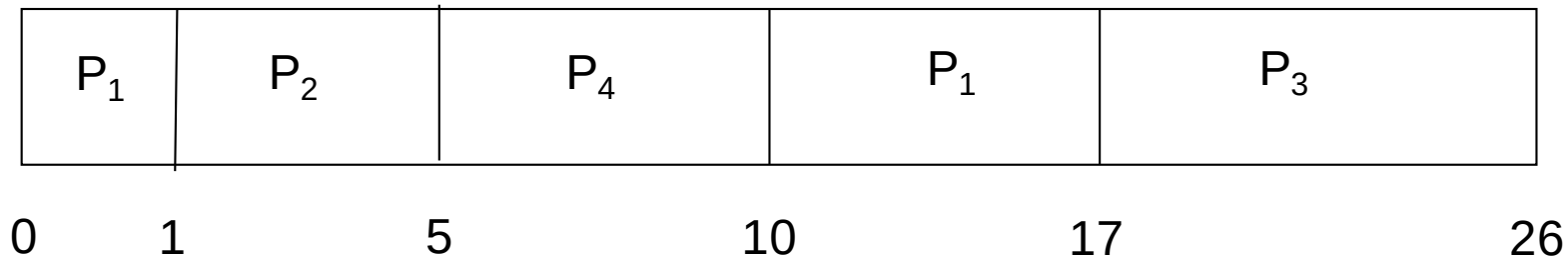
- Shortest remaining time (SRT) is the preemptive version of the SJF algorithm.
- The processor is allocated to the job closest to completion but it can be preempted by a newer ready job with shorter time to completion.
- Impossible to implement in interactive systems where required CPU time is not known.
- It is often used in batch environments where short jobs need to give preference.

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SRT – Shortest Remaining Time

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

- Preemptive SJF Gantt chart



- Average waiting time = $[(10-1)+(1-1)+(17-2)+5-3])/4 = 26/4$
= 6.5 msec

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Priority Scheduling

- A **priority number** (integer) is associated with each process
- The CPU is allocated to the process with **the highest priority** (smallest integer $\equiv$ highest priority)
 - Preemptive
 - Nonpreemptive
- Problem $\equiv$ **Starvation** – low priority processes may never execute
- Solution $\equiv$ **Aging** – as time progresses increase the priority of the process

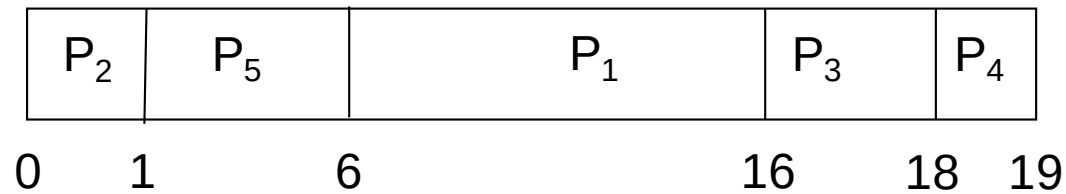
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Priority Scheduling

- Processes with same priority are executed on first come first served basis
- Priority can be decided based on memory requirements, time requirements or any other resource requirement

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	4
P4	1	5
P5	5	2

- Priority scheduling Gantt chart



- Average Waiting time = 8.2 msec



THANK YOU

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